The same year, he visited geologist William Buckland in Scotland to investigate glacial features there, prompting Scottish glaciologist James Forbes to begin similar research in the French Alps.

Some quarters, such as the Catholic Church, still argued that glacial striations had been caused by a great flood or that large silt and rock deposits had been transported by icebergs swept along by the flood. From the 1860s, however, there was wide support for Agassiz's glaciation theory and the idea that glaciers in the Swiss Alps and Norway had once extended much further. It was also accepted that a sheet of ice had once spread across Europe, and south from the Arctic through much of North America, with catastrophic implications for plants and animals.

By the late 1800s and early 1900s, as more expeditions to both Greenland and Antarctica were undertaken, it became known that both areas were still covered in ice. Aerial surveys in the 1920s and 1930s confirmed the extent of their vast ice sheets—now defined as areas of glacier ice exceeding 19,300 sq miles (50,000 sq km); ice caps, such as Iceland's Vatnajökull, are smaller.

Further evidence revealed that there had not been one single ice age, but at least five major ice ages in Earth's long history. The most recent, the Quaternary Ice Age, began 2.58 million years ago and is ongoing. In the last 750,000 years, there have been eight ice advances (glacial periods) and retreats (interglacial periods). During the last glacial period, which ended 10,000–15,000 years ago, ice sheets were up to $2\frac{1}{2}$ miles (4 km) thick, and the sea level was 390 ft (120 m) lower.